



Beyond the numbers: A data analyst journey

Rather than a reading, we invite you to watch [Anna Leach's TEDx talk](#)

so when we're plotting our points in

life either as new parents new graduates

newlyweds we always think that path is

very clear and pretty straight but we

know better we know that it's **windy** and

a little confusing and more like a

journey and an adventure what we're

doing is having experiences that

culminate into the next great thing so

when I've completed my undergrad degree

in math this is how I felt I was

searching for jobs with the title

analytics in it because I knew I wanted

to work with math but I wasn't in the

applied math position those first few

jobs in analytics were really putting

numbers into spreadsheets and then doing

brace yourself really simple formulas in

Excel adding up numbers across the row

and columns the people were great the

companies were nice it just wasn't the

right fit so I kept looking for work and

eventually landed in higher education

now in higher education the process of
onboarding me and getting me started
into work was slightly different here
they had me sit in on meetings learning
about business processes business
processes like how a student would
enroll into a class and how we would
count that enrollment and then I would
sit in meetings learning how to take
that information out of the system that
big black box and put it into a pretty
PDF and send out so that people could
see how many people were enrolled it was

fine it was again good work but I got a

degree in math math to me is is

beautiful take this agave plant from

Tucson where I live it represents to me

a fractal a fractal is a geometric scuse

me

geometry shape that repeats itself and I

know it's not a perfect fractal so I

don't want my undergrad advisor to come

yelling at me but it's still beautiful

math is also certain two plus two equals

four every day it never changes this is

what I love about math so here I am

putting information into spreadsheets no

big deal and again like I said it worked

it wasn't until our system went through

in a complete conversion that it had a

difference in perspective and when I say

a system conversion I'm talking about a

legacy mainframe to a PeopleSoft system

and if you don't know what that means

it's essentially switching from like

Samsung to iPhone or back and forth it's

a completely different way of pulling

out your data out of this big black box

I didn't get drugged into a ton of

meetings cuz I had only been there a
couple years but I did know enough about
critical information that needed to be
reported to the state and federal
government so in our staff meetings we
would talk about how we're going to
report in this new world one of the
critical pieces of information in higher
education is whether or not a student is
a new first-year student freshman if you
will we didn't have a way of recording
this or reporting this in the new system
just yet so somebody had to spend the

time figuring out how to identify these

students I was ready for a new challenge

let's try this out so I approached my

boss and mentor and asked if I could

give it a go and she said go for it so I

met with my colleagues Bob and Harry and

asked them questions

how does this work in the old system how

do you think it's gonna work in the new

system

I spent time face to face with these

people really trying to understand how

the data worked and get that knowledge

from them in a way it was collecting

data myself in a different way then I

had to meet with different departments

because every department also cared

about how this data went into this big

black box or they cared about how it

came out so what I learned in this

portion of data analysis path is that I

had to spend time with people to really

understand the data process next I had

to spend time with the data itself

because as much as I'd love to say that

data goes into a system and it comes out

perfect every time that's

not true there's tons of errors and

quirks and exceptions and timing

differences so I had to spend time with

the data to really understand how it

works

so now let's talk about bias this little

boy is my seven-year-old son he has what

I like to call condiment bias

he has never tried mayonnaise but if he

has to put it on a knife and put it on a

sandwich to help me make lunches for the

week for his dad he holds it as far away

as he possibly can

he's never tried mayonnaise my daughter

on the other hand will only eat ketchup

she has a ketchup bias so the bias that

I brought whenever I was doing data

analysis for a couple the first is that

if something happened in a prior report

in the prior term in the prior year even

the prior week I would assume that I

already analyzed why that happened I'm

not gonna worry about it anymore and

moving on you make a lot of mistakes you

miss a lot of crucial evidence when you

do that the second thing is when I was a
newer analyst especially I was raised to
make yourself irreplaceable the only way
you're gonna keep a job especially in
that job climate that I was in was if
you couldn't be replaced my perception
was that if I looked like I could move
fast do a lot quickly and I knew what I
was talking about even though I didn't I
would not be replaced so when I needed
to spend time face to face with people
and ask questions I was very hesitant I
kept to myself or I did as much research

as I could on my own when all that

knowledge was in the cube right next to

me but I didn't want to look like I

could be replaced

I also made really dumb mistakes because

I was rushing through processes and

reports dumb mistakes like using a

greater than instead of a greater than

or equal to but some people may say so

what well it costs us time the company

time me time other people's time trying

to fix this error trying to explain the

same

and it costs a little bit of your

reputation rushing through and instead

of spending ten more minutes on this

side cost me three more hours on this

side and a little bit of trust so after

you know seven eight years and data

analysis I started learning new patterns

learning new things to try based on what

I'd learned from other people and from

my own experiences so instead of just

diving into a project I take a 30,000

foot view I bother people I look at the

data and I ask questions I look for

patterns and then I start slicing and

dicing so it's at this point in my

career that I thought I had a pretty

clear path I know what's going on I know

where I'm going and I'm pretty good at

this data analysis thing well two years

ago my husband received this

unbelievable offer and my family and I

picked up and moved from Columbus Ohio

to Tucson Arizona it was a big change

but it was an exciting change and we

said to ourselves you know what if we're

gonna do this we're gonna lay the chips

down where we want I wanted to go back

to grad school so why not I'll go back

to grad school I'll work part-time as an

analyst well I was going from full-time

worker and parent and juggling a family

and home to working part-time from home

it's very different it's good but it's

different also a full-time student in an

online program and still a parent and

mom oh and our family and friends were

all 1,700 miles away in Pennsylvania

Ohio oh and I was going into a different

field completely I was going into a

master of learning technologies program

something that had always been a passion

of mine but I just didn't make the time

for my first semester in this master of

long learning technologies program I got

to work with some excellent people the

way that we introduced and met each

other was a little

fun though it was through group work

everybody loves group work we did a

project on evaluating learning

management tool we formed team tank

which is Tim Nunn myself

Natalie Gunter and Karen North and when

our project was completed we spoke with

dr. ana paula crea who was our professor

for the class and she suggested that we

present our project at a conference well

I had never been to a conference let

alone presented at one but this sounded

like okay if they're gonna be with me I

can do this I immediately became a

conference junky I wanted to attend and

present and just be a part of this I

wanted to gather data and it be face to

face with more people I wanted to learn

more experiences and then I wanted to

share what I had to say because after

you present people come up to you

because they have questions and they

want to know it's exciting at that same

conference I met a lady named Katie

Stroud Katie Stroud likes to talk about

the power of story and like any human

being when I heard her information I

related it to myself and I thought

that's what a data analyst is really

you're taking information from this big

black box and you're telling a story

with it based on a question you may be

asked or something that you want to

share

I also learned at these conferences and

in this new world that instructional

designers teachers and educators really

don't have a lot of information or data

when they're trying to make an

assessment of the success of their class

or a tool they're trying to implement or

the teacher themselves they really have

a hard time measuring that qualitative

data I really began to appreciate the

fact that I had these massive data

points in my comfort zone job and then I

started thinking that maybe data

analysis is something that we should all

start to embrace we should all start to

try and figure out where we can pull

this information from at another

conference that was more academic based

I met a lot of graduate students and

professors that were presenting on their

research remember that I'm new to the

world of graduate school so when I would

listen to their presentation afterwards

I would say things like how it's

wonderful so what's next what are you

gonna do with this

everybody's response was to get

published and I thought well that that's

awesome let's do this get published

thing but what else are you going to do

with it what problem are you trying to

solve who are you trying to help what

are you going to look into next and I

realized I was coming across a bit of a

aggressive and backed off and just

stopped asking and started listening one

presentation by a dr. Thomas Reeves

started talking about a rift between

educational practice and academic

research the teachers feel like

researchers aren't looking into the

things that they need or sharing

information with them and then

researchers feel like they provide all

this information and do all this

research but it's not applied and that's

when I realized that I have a different

perspective I'd been a data analyst all

this time and these pressures from

graduate school are completely different

and I thought again there's a gap here

this is data analysis this is one system

in one system and nothing's happening in

between this is where I started to think

there's purpose in data analysis to

bring these two people together to bring

these groups together to find a way to

share the information so when you think

about data analysis think about spending

time with people think about asking

questions thinking about lifting the

heavy rock before the presentation look

underneath things really appreciate the

time that can be spent and the data

that's out there data analysis is as

much an art as it is a science anyone

from any background can really

appreciate its beauty

and anyone from any background can

really appreciate the power it can bring

to our relationships thank you.